

Sahil

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GitHub Profile
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Professional Summary

A Cybersecurity and Computer Science student with hands-on experience in network security, machine learning and full-stack development. Proven experience in developing real-time security solutions such as Intrusion detection system, packet sniffer, network traffic analyzers by using Python, Scapy and machine learning algorithms. Experience in developing reactive websites by using HTML, CSS, JavaScript, React and Node.js. Good understanding in cybersecurity, digital forensics and software engineering and interested in solving real-world security problems with practice and research oriented projects. Fast learning ability with problem solving skills, teamwork attitude, and enthusiasm towards the future technologies and security with in cybersecurity.

EDUCATION

- National Forensic Sciences University** 2022
BTech - MTech(Integrated) Computer Science And Engineering Specialization in Cyber Security CGPA: 8.80
- R K C Vidya Vihar, Rewari, Haryana** 2022
Class – 12 (Central Board of Secondary Education) Percentage: 92%

EXPERIENCE

- RailTel Corporation of India Ltd** May 2026 - July 2026
Security Operation Center Analyst On-Site
 - Monitoring enterprise-scale network infrastructure of a Government of India telecom company in real time using SIEM platforms and Elastic Stack (ELK) to detect anomalies, security events, and potential threats across production environments.
 - Performing log ingestion, parsing, and correlation across multiple data sources to identify Indicators of Compromise (IoCs) and support proactive threat hunting workflows.
 - Supporting incident detection and response operations by triaging security alerts, classifying threats by severity, and escalating confirmed incidents through defined SOC response procedures.
- IFSO/NCFL, Special Cell, Delhi Police** May 2025 - July 2025
Network Security Intern On-Site
 - Designed and implemented an Advanced Packet Sniffer & Intrusion Detection System supporting IPv4/IPv6, with real-time threat detection, PCAP analysis and firewall integration using iptables.
 - Gained hands-on experience with industry-standard forensic tools including Cellebrite UFED, FTK, Falcon, and Magnet AXIOM for digital evidence extraction and analysis.
- Prodigy Info Tech** October 2024 - November 2024
Machine Learning Intern Remote
 - Project Development & Implementation: Worked on distinct tasks as part of a month-long project, applying machine learning techniques including supervised and unsupervised learning, data preprocessing, and model evaluation to solve real-world business problems.
 - Model Optimization and performance tuning: Contributed to improving the performance of machine learning models by experimenting with various algorithms, hyperparameter tuning, and evaluating the accuracy, precision, and recall of models using cross-validation.

PROJECTS

- BLE-SPATIAL: Bluetooth-Based Invisible User Interface for Ambient Spatial Computing**
- Passive RF Signal Intelligence:** Built a 7-phase BLE RSSI pipeline that passively fingerprints user presence and movement using only ambient Bluetooth radio signals — no camera, microphone, or GPS required, demonstrating a privacy-preserving alternative to traditional surveillance-based sensing.
- Signal Processing & Anomaly Detection:** Implemented a dual-stage noise reduction pipeline (moving average + 1D Kalman filter, $Q=0.008$, $R=2.0$) on raw BLE RSSI streams to isolate meaningful signal trends from environmental interference, applying the same filtering principles used in network anomaly detection.
- ML-Powered Gesture Classification:** Trained a Random Forest classifier (100 trees, 99.17% test accuracy) on 13 extracted RSSI features per sample to classify 4 gesture types in real time, using triple-confirmation logic at $\geq 70\%$ confidence threshold to eliminate false positives.
- Context-Aware Security Automation:** Designed a rule-based context engine with a safety guard module that enforces pre-execution confidence checks and cooldown policies before triggering OS-level actions (screen lock via ctypes, volume control via pycaw), preventing unauthorized or accidental system access.
- Stack:** Python, Bleak (BLE), NumPy, SciPy, scikit-learn, PyQt6, pycaw, PyAutoGUI, ctypes, asyncio, SQLite (WAL mode).

- **VMGUARD: Hypervisor-Level Agentless Behavioral Monitoring System**
- Agentless Architecture: Built a host-based VM security monitoring platform that observes a Windows VM entirely from the host machine with zero software, agent, or driver installed inside the guest VM.
- Multi-Channel Monitoring: Simultaneously monitored three external channels — network traffic via Scapy, shared folder file events via Watchdog with SHA-256 and Shannon entropy analysis, and host resource metrics via psutil.
- Behavioral Threat Scoring: Implemented a rule-based Behavioral Analysis Engine that correlates signals from all three channels to calculate a composite threat score (0–100%) and detect attack patterns like C2 beaconing, malware drops, and data exfiltration.
- Forensic Analysis: Developed functionality to upload and analyze.pcap files for post-incident security investigations.
- Stack: Python, FastAPI (22+ REST endpoints), Streamlit, Plotly, Scapy, Watchdog, psutil, SQLite (WAL mode).
- **NETSECURE: Unified Platform for Real-Time NIDS**
- Live Capture: Utilized Scapy and Python threading to capture and analyze live network traffic without freezing the UI.
- Intrusion Detection: Implemented a dual-engine Intrusion Detection System with both signature-based and behavioral analysis to identify suspicious activities.
- Dynamic Blocking: Integrated with system firewalls (iptables / netsh) via subprocess to automatically block malicious IP addresses.
- Forensic Analysis: Developed functionality to upload and analyze.pcap files for post-incident security investigations.
- Stack: Built with a Python Flask backend and a JavaScript frontend, communicating via a REST API.
- **Network Packet Analyzer**
- Traffic Analysis: Captured and displayed essential packet details including source/destination IP, protocol, and port numbers.
- PCAP File Management: Enabled saving live captured traffic to .pcap files and importing existing .pcap files for offline analysis.
- Data Visualization: Created pie charts for protocol distribution and bar graphs for traffic volume per IP, providing clear insights into network activity.
- Stack: Engineered with a Python backend and a React.js / Node.js frontend, connected via WebSockets for real-time updates.

Skills

- Technical Skills:

Tools and platforms – Scapy, Wireshark, Cellebrite UFED, FTK, Magnet AXIOM, Burp Suite, Falcon, iptables, Nmap, Git.

Programming – Python, C, C++, Java(basic).

Cybersecurity and SOC – SIEM fundamentals, Log analysis, Security monitoring, threat detection concepts, Application security.

Web & Software Development – Flask, React.js, Node.js, REST APIs, WebSockets, HTML, CSS, JavaScript.

Databases and Systems – MySQL, Linux fundamentals.

Networking – Network Security, Intrusion Detection Systems (IDS), Packet Sniffing, Packet Analysis, PCAP Analysis, Threat Detection, Firewall Integration, Network Monitoring, Traffic Analysis, IPv4/IPv6 Analysis, Cybersecurity Fundamentals, Digital Forensics, Mobile Forensics.

- Soft Skills:

Problem Solving, Leadership, Communication, Team Collaboration, Event Management, Analytical Thinking.